Saliency Detection by Superpixel-Based Sparse Representation

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Abstract. In this paper, we propose a novel model for predicting visual saliency by superpixel-based sparse representation. A superpixel-based sparse representation utilizes the Simultaneous Orthogonal Matching Pursuit algorithm to extract the sparse features from color maps and activation maps of complex cells. The saliency is calculated according to the sparse features from different dictionaries. To guarantee the robustness of the proposed method, the proposed method is performed on a multi-scale basis thus the final saliency result is obtained by using the saliency maps from different scales. Experimental results on multiple datasets show that the proposed model outperforms several advanced methods for saliency prediction.

Keywords: Saliency prediction · Superpixel · Sparse representation

1 Introduction

The Human Vision System (HVS) can quickly prioritize external visual stimuli and pick out interesting parts from a scene. Simulating such human capability with computer has been a topic of interest for researchers in the field of computer vision over the past few decades. Saliency detection aims at highlighting visually salient regions or objects in an image, and results of it can be used to facilitate other computer vision tasks such as image segmentation [5], object detection [14], person re-identification [25], etc. Due to the importance of saliency detection, a large number of models have been proposed to capture different saliency cues. Saliency detection models can be categorized as bottom-up and top-down approaches. Bottom-up methods are data-driven, fast and pre-attentive, while top-down methods are goal-driven, slow and entail supervised learning with class labels. In this paper, we focus on the bottom-up method for images saliency detection.

Current studies reveal that the sparse representation of image statistics exists in the primate visual system [16]. These studies in neuroscience suggest that sparse coding strategy is an efficient way for modelling natural scene as sparse representation, which simulates the V1 population responses to natural stimuli. Extracting the independent sparse features from images has been demonstrated

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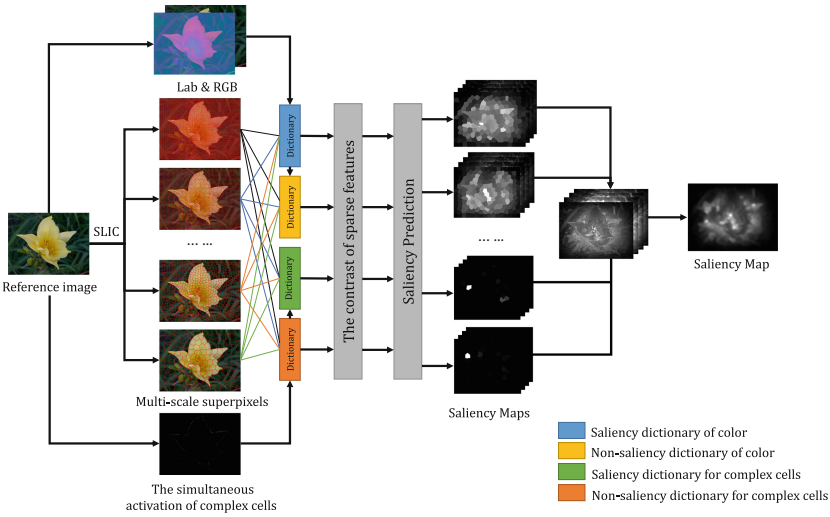

Fig. 1. Framework overview of the proposed model. The model utilises the Simultaneous Orthogonal Matching Pursuit algorithm to extract the sparse features from color maps and complex cells' activation map. Then, the saliency map is formed by the contrast of sparse features under different scales and different dictionaries. (Color figure online)

to be highly effective for many computer vision tasks, leading to better performance for saliency detection.

Saliency models based on sparse representation [6] have been proposed to extract the saliency map of the input image by computing the center-surround differences between image patches mainly. However, these models ignore the fact that image patches can span multiple distinct image regions. Image patches are local regions with fixed size and shape. They may not effectively exploit spatial information of the image thus leading to lower accuracy in saliency detection. Compared with image patches, superpixels are obtained from an over-segmented image, and they aggregate visually homogeneous pixels while respecting natural boundaries. Therefore, superpixels are superior to patches in saliency detection and superpixel-based sparse representation are more likely to simulate the V1 population responses to natural stimuli. Zhang et al. [24] and Li et al. [12] have considered superpixels and sparse representation. They all utilize data-specific features of superpixels to build a dictionary. But, superpixel is a small spatial region whose shape and size can be adaptively changed for different spatial structures. These methods may lose detailed features of superpixels using data-specific features.

To overcome the above shortcomings, we propose a novel sparse representation based on superpixels. The simultaneous orthogonal matching pursuit algorithm [19] is utilized to extract the sparse features of pixels in a superpixel by the joint sparse regularization. We also apply this method to construct a multi-layer sparse

coding network [8] based on the theory argued in [9]. The framework overview of the proposed model is shown in Fig. 1. The proposed model firstly extracts sparse features of multi-scale superpixels, which have been proved to be an important cue to improve the saliency region detection in [18], from the color map and complex cells responses. Then, the model measures saliency by calculating the contrast of sparse features over the entire image, combining saliency maps from multi-scale superpixels. To evaluate the performance of the proposed model, we apply it on several datasets for eye fixation prediction. The experimental results show the better performance of the proposed saliency detection model over other existing ones.

The rest of this paper is organized as follows. In Sect. 2, the adopted perceptual domain knowledge for the proposed model are briefly reviewed. Section 3 introduces the proposed framework for saliency detection. Experimental results on two benchmark datasets are presented in Sect. 4. The final section concludes this paper and suggests future research directions.

2 Preliminaries

2.1 Sparse Representation

Sparse representation model is a linear model, where the original input signal can be represented as a linear combination of basis vectors. Here, the response is sparse to stimulate the basis vectors of the input image and shows the same characteristics of the human visual system. Formally, sparse representation model depicts signals by the linear combination of a few items from a pre-specified dictionary. The performance of the sparse representation relies heavily on the quality of the dictionary D. Recently, many dictionary learning algorithms have been proposed to achieve compact representation. One widely used dictionary learning algorithm is the K-SVD [2].

Given an over-complete dictionary $D = [d_1, d_2, ..., d_n] \in R^m$, where normally $m \leq n$, n represents the size of dictionary, and an input signal $I \in R^m$, m is the dimension of the signal. Then the sparse coding coefficients α for x is obtained by solving:

$$\min_{\alpha} \|\alpha\|_0 \;\; s.t. \;\; \|I - D\alpha\|_2^2 \leq \epsilon \tag{1}$$

The above model aims to seek the most compact representation for the input signal x based on the given dictionary D. However, the optimization for (1) is NP-hard combinatorial optimization problem which can be solved with Orthogonal Matching Pursuit (OMP) algorithm [15].

Unlike traditional saliency models based on sparse representation utilize sparse features to represent local regions of fixed size and shape, the proposed model can effectively capture the correlations of each superpixel by utilizing the joint sparse representation. This model is further described in the following sections.

2.2 Multi-layer Sparse Coding Model

The spatial classical receptive fields (CRFs) of neurons in the primary visual cortex (V1) of primates are selective for location, orientation, and frequency [9]. Hubel and Wiesel in [9] argued that most neurons can be grouped into two categories: simple cells and complex cells. They have different responses to the outside signal stimulation. Simple cells have strong and linear responses to the specific characteristics, such as location, orientation, and frequency, while complex cells are not sensitive to the outside stimuli because of their larger receptive field area. Their responses have nonlinear characteristics. Inspired by this, Hoyer proposed a multi-layer sparse coding model [8] to describe the response properties of high-level neurons further.

To model the V1 complex cells responses, the multi-layer sparse coding [8] takes squares and sums up the linear filtering results of the two quadrature Gabor filters. The output of a model complex cell can be described as follows:

$$C_{\{x_c,y_c,\theta\}} = (\sum_{x,y} G_{\{e,x_c,y_c,\theta\}} I(x,y))^2 + (\sum_{x,y} G_{\{o,x_c,y_c,\theta\}} I(x,y))^2 \tag{2}$$

when fed with the image $I(x,y)$. Here $G_{\{e,x_c,y_c,\theta\}}$ and $G_{\{o,x_c,y_c,\theta\}}$ are even- and odd-symmetric (respectively) Gabor filters, centered on (x_c, y_c) and of orientation θ. In our model, we utilize this method to construct complex cells' activation maps of images.

3 Saliency Detection by Superpixel-Based Sparse Representation

As is shown in Sect. 2.2, simple cells and complex cells are sensitive to the specific characteristics, such as location, orientation, and frequency, but they are not sensitive to the color. However, color is an important factor influencing people to pick out interesting parts from a scene. Apart from response maps of complex cells, we also choose both RGB and Lab color spaces as color descriptors.

As the overview of the proposed saliency detection model shown in Fig. 1, the SLIC algorithm [1] is used, and the image is converted into the RGB and Lab color space to segment the original image into multi-scale superpixels. Non-saliency regions can be represented by a sparsely coded dictionary [13]. To maximize the difference between sparse coefficients of superpixels in saliency regions and non-saliency regions, we define four over-complete dictionaries trained by the K-SVD algorithm [2] on FIGRIM Fixation Dataset [4]. Half of this dictionary belongs to saliency dictionaries which are regarded as the potential saliency regions. The other half belongs to the non-saliency dictionary. Two saliency dictionaries correspond to color features and complex cells responses (respectively), and non-saliency dictionaries have the same classification. Then, the sparse features of each superpixel are captured based on the color and complex cells responses as follows:

3.1 Superpixel-Based Sparse Representation

Superpixel is a perceptually meaningful atomic region by grouping pixels and can be used to replace the rigid structure of the pixel grid. However, superpixels' shape and size can be adaptively changed for different spatial structures. Each superpixel is composed of some pixels $[p_{i,1}, p_{i,2}, ...]$, and pixels within each superpixel are assumed to have very similar characteristics. Based on this theory, superpixel-based sparse representation is similar to simultaneous sparse approximation problem.

A simultaneous sparse approximation problem can be described as:

Given several input signals, approximate all these signals at once using different linear combinations of the same elementary signals, while balancing the error in approximating the data against the total number of elementary signals that are used [19].

As described above, a superpixel can be treated as a collection of multiple signals, where the signal is equivalent to a pixel in an image. To solve the simultaneous sparse approximation problem, the Simultaneous Orthogonal Matching Pursuit algorithm is proposed in [19]. Given a set of signals $X = [X^1, \ldots, X^n]$ in $R^{m\times N}$, where the X^i are in $R^{m\times n_i}$, and an over-complete dictionary D in $R^{m\times p}$, the Simultaneous Orthogonal Matching Pursuit algorithm solves a matrix of coefficients $A = [A^1, \ldots, A^n]$ in $R^{p\times N}$ which is an approximate solution of the following NP-hard problem

$$\forall i \quad \min_{A^i \in R^{p\times n_i}} \|X^i - DA^i\|_F^2 \ s.t. \ \|A^i\|_{0,\infty} \leq L. \tag{3}$$

For superpixels-based sparse representation, it can be described as follows:

Each pixel is seen as a signal, then pixels within each superpixel can be arranged into a matrix S_i and are assumed to have very similar characteristics. Pixels within each superpixel S_i shall be simultaneously decomposed by linear combinations of a few common atoms from the dictionary, which can be seen as the joint sparse regularization. Let a matrix A_i be the sparse coefficients of the S_i. The joint sparse matrix A_i is an approximate solution of the following optimization problem:

$$\forall i \quad \min_{A^i \in R^{p\times n_i}} \|S_i - DA_i\|_2 \ s.t. \ \|A_i\|_0 \leq K. \tag{4}$$

where A_i is the sparse coefficients of the superpixel S_i and D is the over-complete dictionary utilizing the method mentioned above. Here, the superpixel-based sparse representation is described a simultaneous sparse approximation problem and the Simultaneous Orthogonal Matching Pursuit algorithm is used to solve the problem above efficiently. The solving process is shown in as Algorithm 1. The output of the Simultaneous Orthogonal Matching Pursuit algorithm, A set Λ_t containing T indices, determines the sparse coefficients A_i of the superpixel S_i.

Pixels within each superpixel are assumed to have very similar characteristics, so their sparse coefficients are also similar. In our model, we use the means of the joint sparse coefficients A_i as the sparse coefficient α_i of the superpixel S_i.

Algorithm 1. Simultaneous Orthogonal Matching Pursuit

Input: – A $m * N$ signal matrix S,
– The stopping criterion and the reconstruction error ϵ,
– An over-complete dictionary D.

Output: A set Λ_t containing T indices, where T is the number of iterations completed.

1: Initialize the residual matrix $R_0 = S$, the index set $\Lambda_0 = \emptyset$, and the iteration counter $t = 1$.
2: Find an index λ_t that solves the easy optimization problem

$$\max_{\omega \in \Omega} \sum_{n=1}^{N} |\langle R_{t-1} e_n, d_\omega \rangle| .$$

e_k denotes the kth canonical basis vector in R_{t-1} and d_ω is ωth column of the over-completed dictionary D, where the parameter ω ranges over an index set Ω.
3: Set $\Lambda_t = \Lambda_{t-1} \cup \{\lambda_t\}$.
4: Determine the orthogonal projector P_t onto the span of the atoms indexed in Λ_t.
5: Calculate the new approximation and residual:

$$R_t = S - P_t S.$$

6: Increment t, and return to Step 2 unless the stopping criterion is met.

In this way, each superpixel S_i is turned to be a linear combination of basis vectors with a sparse coefficients α_i. And this is used to be the feature vector in saliency detection.

3.2 Saliency Calculation Based on Sparse Features

The analysis noted that a superpixel with much difference compared with the others has a higher probability of attracting human's attention. Spatial relationship plays an important role. The closer the distance, the stronger effect between superpixels. Therefore, the Gaussian function can be a good choice to describe this:

$$G_\delta(\|a - b\|) = e^{-\frac{\|a-b\|^2}{\delta^2}} \tag{5}$$

Human eye-tracking studies have shown that gaze fixations are biased toward the center of natural scene stimuli [20]. Center bias has been much discussed in the saliency literature. Here, center bias appears more directly as a factor that influences the relative weights assigned to the score of saliency defined by local superpixels. This effectively means that superpixels closer to the center have more influence in determining saliency value.

Obtain the saliency of each superpixel by calculating the sum of sparse features difference between the superpixel itself and other superpixels. This process can be described as follows:

$$Sal_i^{SP} = \sum_{j=1}^{n} D_{i,j} G_\delta(\|P_i - P_j\|) G_{\delta_c}(\|P_i - P_c\|) \tag{6}$$

where, $D_{i,j}$ is the superpixel difference between superpixels i and j, c is the spatial center of the image, G_{δ_c} is a Gaussian function which controls the amount of center bias based on δ_c, G_δ is a Gaussian function which controls the influence of the distance between the superpixel itself and other superpixels based on δ, Sal_i^{SP} is the saliency value of superpixel i. Here we use the distance between superpixels as the weight of the superpixels difference to calculate the saliency map. We compute $D_{i,j}$ based on sparse features α as follows:

$$D_{i,j} = \frac{\alpha(i) \cdot \alpha(j)}{\|\alpha(i)\| \cdot \|\alpha(j)\|} \tag{7}$$

Every sparse feature is based on the same basis, so the sparse features can be taken as sparse vectors based on same basis vector. Cosine similarity is easy to interpret and simple to compute for sparse vectors. It always performs better for the contrast of sparse vectors. Therefore, we utilize cosine similarity to improve the accuracy of saliency maps.

Combining the saliency map with four sparse responses by four different dictionaries and several scales as computed above. We calculate the average saliency under different scales as saliency value of one pixel:

$$Sal_i^P = \frac{1}{m} \sum_{s=1}^{m} \sum_{d=1}^{4} Sal_{i,r,d}^{SP} \tag{8}$$

where, $Sal_{i,r}^{SP}$ is the saliency value of the superpixel where pixel i is located at scale s by dictionary d, m is the number of superpixels scales.

Implementation Details. Each input image is first converted into the RGB and Lab color space and segmented into superpixels by four scales. Then, get complex cells' activation map of the input image by multi-layer sparse coding network [8]. The Gaussian envelope had an aspect ratio of 1.5, and the spatial frequency of the sinusoid was chosen to give 2–3 main inhibitory regions. Four activation values of four complex cells with four orientations is extracted from every texel. These values construct complex cells' activation map of the input image. We define four over-complete dictionaries trained by the K-SVD algorithm [2] on FIGRIM fixation dataset [4]. FIGRIM fixation dataset consists of 2787 natural scenes from 21 different indoor and outdoor scene categories. After superpixel-based sparse representation for color maps and activation map, the model combines the saliency map with four sparse responses by four different dictionaries and several scales. Before making the summation, each saliency map is normalized.

4 Experimental Results

In this section, the performance of the model we proposed is evaluated in MIT1003 dataset [11] and Toronto dataset [3]. The MIT1003 dataset [11] consists of 1003 natural indoor and outdoor scenes. The eye tracking data in MIT1003

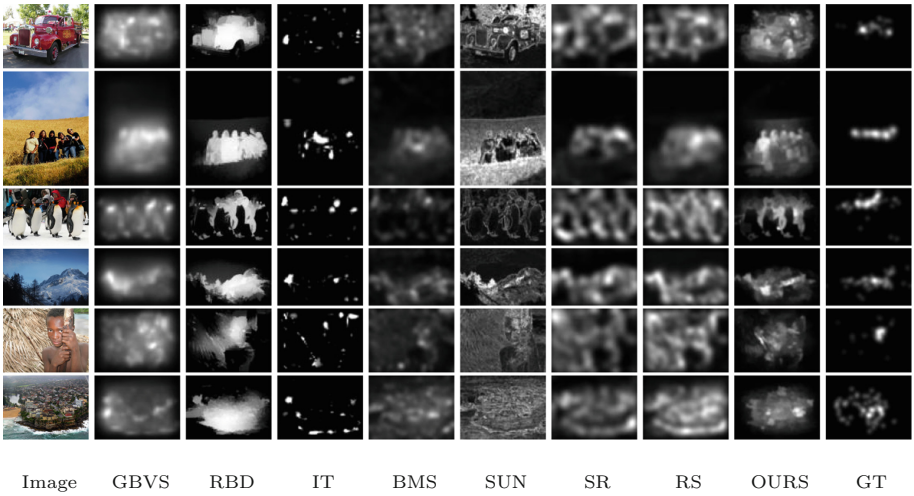

Fig. 2. Comparison between our model with the state-of-art fixation prediction models.

Table 1. Comparison results of Similarity, CC, AUC-Judd and AUC-Borji on the MIT1003 dataset. ***Scores in bold means the best performance.***

Metrice	Toronto DataSet				MIT1003 DataSet			
	Similarity	CC	AUC-Judd	AUC-Borji	Similarity	CC	AUC-Judd	AUC-Borji
GBVS [17]	0.466	0.527	0.819	0.807	0.343	0.380	0.806	0.796
RBD [26]	0.465	0.480	0.798	0.750	0.376	0.382	0.801	0.774
IT [10]	0.415	0.383	0.764	0.752	0.294	0.252	0.721	0.713
BMS [22]	0.401	0.356	0.720	0.675	0.326	0.291	0.744	0.701
SUN [23]	0.349	0.216	0.671	0.652	0.264	0.188	0.667	0.653
SR [7]	0.288	0.157	0.661	0.531	0.236	0.129	0.645	0.534
RS [21]	0.478	0.510	0.815	0.797	0.354	0.386	0.799	0.785
Proposed	**0.527**	**0.596**	**0.836**	**0.816**	**0.402**	**0.443**	**0.826**	**0.808**

dataset is collected using a head-mounted eye tracking device for 15 different viewers. The Toronto dataset [3] consists of 120 color images of outdoor and indoor scenes. A large portion of images in Toronto dataset do not contain particular regions of interest. The proposed saliency model is compared with several eye fixation prediction methods GBVS [17], RBD [26], IT [10], BMS [22], SUN [23], SR [7] and RS [21].

For an exhaustive performance evaluation, almost all of the commonly used evaluation metrics, including AUC(Area Under the Receiver Operation Characteristics Curve), CC(Correlation Coefficient) and similarity, are adopted in this paper. Among these evaluation metrics, AUC, containing two accepted version: the Borja's version and Judd's version, is the most widely used metric and is a

comparison between saliency map and all eye fixation points. While CC and similarity are the comparisons between saliency map and eye fixation density map.

As shown in Table 1, the proposed saliency model achieves the best performance on CC, Similarity, AUC_Borja and AUC_Judd evaluation metrics on the Toronto dataset. On the MIT1003 dataset, the proposed saliency model also outperforms all of the evaluation metrics. Analyze the results shown in Table 1 and our model extracts from the better sparse features of superpixels, and cosine similarity can better measure the difference between sparse features. The multi-scale superpixel is also proved to be an important cue to improve the saliency region detection. Samples of fixation prediction results in this two experimental datasets are shown in Fig. 2.

5 Conclusions

In this paper, we propose a novel saliency detection model based on superpixel-based sparse representation. The novel method of superpixel-based sparse representation utilizes the Simultaneous Orthogonal Matching Pursuit algorithm to extract the sparse features from color maps and complex cells' activation map. Then, the saliency map is formed by the contrast of sparse features under different scales and different dictionaries. Experimental results show that our approach is effective when compared to the several advanced methods.

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References

1. Achanta, R., Shaji, A., Smith, K., Lucchi, A., Fua, P., Süsstrunk, S.: Slic superpixels compared to state-of-the-art superpixel methods. IEEE Trans. Pattern Anal. Mach. Intell. **34**(11), 2274–2282 (2012)
2. Aharon, M., Elad, M., Bruckstein, A.: *rmk*-svd: an algorithm for designing overcomplete dictionaries for sparse representation. IEEE Trans. Signal Process. **54**(11), 4311–4322 (2006)
3. Bruce, N.D.B., Tsotsos, J.K.: Saliency based on information maximization. Adv. Neural Inf. Process. Syst. **18**(3), 298–308 (2005)
4. Bylinskii, Z., Isola, P., Bainbridge, C., Torralba, A., Oliva, A.: Intrinsic and extrinsic effects on image memorability. Vision. Res. **116**, 165–178 (2015)
5. Donoser, M., Urschler, M., Hirzer, M., Bischof, H.: Saliency driven total variation segmentation. In: IEEE International Conference on Computer Vision, pp. 817–824 (2009)
6. Fang, Y., Lin, W., Fang, Z., Chen, Z., Lin, C.W., Deng, C.: Visual acuity inspired saliency detection by using sparse features. Inf. Sci. **309**, 1–10 (2015)
7. Hou, X., Zhang, L.: Saliency detection: a spectral residual approach. In: IEEE Computer Society Conference on Computer Vision and Pattern Recognition, pp. 1–8 (2007)
8. Hoyer, P.O., Hyvärinen, A.: A multi-layer sparse coding network learns contour coding from natural images. Vision. Res. **42**(12), 1593–1605 (2002)

9. Hubel, D.H., Wiesel, T.N.: Receptive fields and functional architecture of monkey striate cortex. J. Physiol. **195**(1), 215–243 (1968)
10. Itti, L., Koch, C., Niebur, E.: A model of saliency-based visual attention for rapid scene analysis. IEEE Trans. Pattern Anal. Mach. Intell. **20**(11), 1254–1259 (1998)
11. Judd, T., Ehinger, K., Durand, F., Torralba, A.: Learning to predict where humans look. In: IEEE International Conference on Computer Vision (2009)
12. Li, N., Sun, B., Yu, J.: A weighted sparse coding framework for saliency detection. In: IEEE Conference on Computer Vision and Pattern Recognition, pp. 5216–5223 (2015)
13. Li, X., Lu, H., Zhang, L., Ruan, X., Yang, M.H.: Saliency detection via dense and sparse reconstruction. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 2976–2983 (2013)
14. Luo, P., Tian, Y., Wang, X., Tang, X.: Switchable deep network for pedestrian detection. In: IEEE Conference on Computer Vision and Pattern Recognition, pp. 899–906 (2014)
15. Mallat, S.G., Zhang, Z.: Matching pursuits with time-frequency dictionaries. IEEE Trans. Signal Process. **41**(12), 3397–3415 (1993)
16. Olshausen, B.A., Field, D.J.: Emergence of simple-cell receptive field properties by learning a sparse code for natural images. Nature **381**(6583), 607 (1996)
17. Schlkopf, B., Platt, J., Hofmann, T.: Graph-based visual saliency. Adv. Neural Inf. Process. Syst. **19**, 545–552 (2006)
18. Tong, N., Lu, H., Zhang, L., Xiang, R.: Saliency detection with multi-scale superpixels. IEEE Signal Process. Lett. **21**(9), 1035–1039 (2014)
19. Tropp, J.A., Gilbert, A.C., Strauss, M.J.: Algorithms for simultaneous sparse approximation. Part I: greedy pursuit. Sig. Process. **86**(3), 572–588 (2006)
20. Tseng, P.H., Carmi, R., Cameron, I.G., Munoz, D.P., Itti, L.: Quantifying center bias of observers in free viewing of dynamic natural scenes. J. Vis. **9**(7), 4 (2009)
21. Yan, J., Liu, J., Li, Y., Niu, Z., Liu, Y.: Visual saliency detection via rank-sparsity decomposition. In: IEEE International Conference on Image Processing, pp. 1089–1092. IEEE (2010)
22. Zhang, J., Sclaroff, S.: Saliency detection: a boolean map approach. In: IEEE International Conference on Computer Vision, pp. 153–160 (2013)
23. Zhang, L., Tong, M.H., Marks, T.K., Shan, H., Cottrell, G.W.: SUN: a Bayesian framework for saliency using natural statistics. J. Vis. **8**(7), 32.1–32.20 (2008)
24. Zhang, L., Zhao, S., Liu, W., Lu, H.: Saliency detection via sparse reconstruction and joint label inference in multiple features. Neurocomputing **155**, 1–11 (2015)
25. Zhao, R., Ouyang, W., Wang, X.: Unsupervised salience learning for person re-identification. In: IEEE Conference on Computer Vision and Pattern Recognition, pp. 3586–3593 (2013)
26. Zhu, W., Liang, S., Wei, Y., Sun, J.: Saliency optimization from robust background detection. In: IEEE Conference on Computer Vision and Pattern Recognition, pp. 2814–2821 (2014)